Rolling Maximum-Sharpe Multi-Asset Portfolio

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Project Overview

This project develops a **rolling maximum-Sharpe asset-allocation strategy** across six exchange-traded funds:

Ticker	Portfolio role
VOO	U.S. large-cap equities
VXUS	International equities outside the United States
TLT	Long-term U.S. Treasury bonds
TIP	U.S. inflation-protected Treasury bonds
GLD	Gold exposure
BIL	Short-term U.S. Treasury bills

The strategy uses historical returns, volatilities, and correlations to estimate the long-only portfolio allocation with the highest expected Sharpe ratio. Portfolio weights are reoptimized every 10 trading days using a rolling three-year estimation window.

The strategy is evaluated on a chronological out-of-sample period and compared with a passive investment in VOO.

Strategy Design

Portfolio objective

At every rebalancing date, the optimizer selects portfolio weights that maximize the estimated annualized Sharpe ratio:

$$\text{Sharpe Ratio} = \frac{E(R_p) - R_f}{\sigma_p}$$

where:

- $E(R_p)$ is the portfolio's estimated annual return
- R_f is the annualized three-month Treasury bill rate
- σ_p is the portfolio's estimated annual volatility

The optimization is implemented by minimizing the negative Sharpe ratio using the **Sequential Least Squares Programming**, or SLSQP, algorithm.

Portfolio return and risk

The expected portfolio return is:

$$E(R_p) = w^{\top}\mu$$

Portfolio volatility is:

$$\sigma_p = \sqrt{w^{\top}\Sigma w}$$

where:

- w is the vector of portfolio weights
- μ is the vector of annualized historical mean returns

- $\Sigma$is the annualized covariance matrix

The covariance matrix allows the model to account for the interaction between assets rather than evaluating each asset independently.

Backtest Configuration

Setting	Value
Starting date	1 January 2018
Assets	VOO, VXUS, TLT, TIP, GLD and BIL
Training proportion	70%
Out-of-sample proportion	30%
Rolling lookback	756 trading days
Approximate lookback length	3 years
Reoptimization frequency	Every 10 trading days
Approximate rebalance frequency	Every 2 weeks
Minimum weight per asset	0%
Maximum weight per asset	100%
Net-weight constraint	100%
Risk-free-rate proxy	13-week U.S. Treasury bill yield, ^IRX
Benchmark	VOO
Optimizer	SLSQP

The data split is chronological. The first 70% of observations establishes the beginning of the out-of-sample period, while the final 30% is used for strategy evaluation.

At each out-of-sample rebalancing date, the optimizer uses only information available before that date:

Python

```
available_returns = (
returns.loc[returns.index < date]
.tail(lookback_days)
)
```

Previously observed out-of-sample returns may enter later rolling windows because those returns would already be publicly known by the later rebalancing date. Future observations are not used.

Walk-Forward Process

The rolling model follows the process below:

1. Select the most recent 756 trading days before the rebalancing date.
2. Estimate annualized asset returns.
3. Estimate the annualized covariance matrix.
4. Estimate the annualized risk-free rate.
5. Find the weights that maximize the estimated Sharpe ratio.
6. Apply the optimized weights to subsequent daily returns.
7. Allow the weights to drift with asset performance between rebalancing dates.
8. Reoptimize after 10 trading days.
9. Combine all realized daily returns to evaluate out-of-sample performance.

The daily weight-drift formula is:

$$w_{i,t+1} = \frac{w_{i,t}(1 + R_{i,t+1})}{1 + R_{p,t+1}}$$

This formula updates each asset's weight according to the change in its market value relative to the total portfolio value.

Out-of-Sample Results

Performance measure	Optimized portfolio	VOO
Realized annualized Sharpe ratio	1.7911	0.7928
Approximate ending value of $1 from chart	$1.97	$1.49

$$1.7911 - 0.7928 = 0.9983$$

$$\frac{1.7928}{0.7928} - 1 \approx 125.9\%$$

The long-only optimized portfolio generated a realized annualized Sharpe ratio of **1.7911**, compared with **0.7928** for VOO. The optimized portfolio therefore produced a Sharpe ratio **0.9983 points**, or approximately **125.9%**, higher than the benchmark during the selected out-of-sample period.

Portfolio-Growth Interpretation

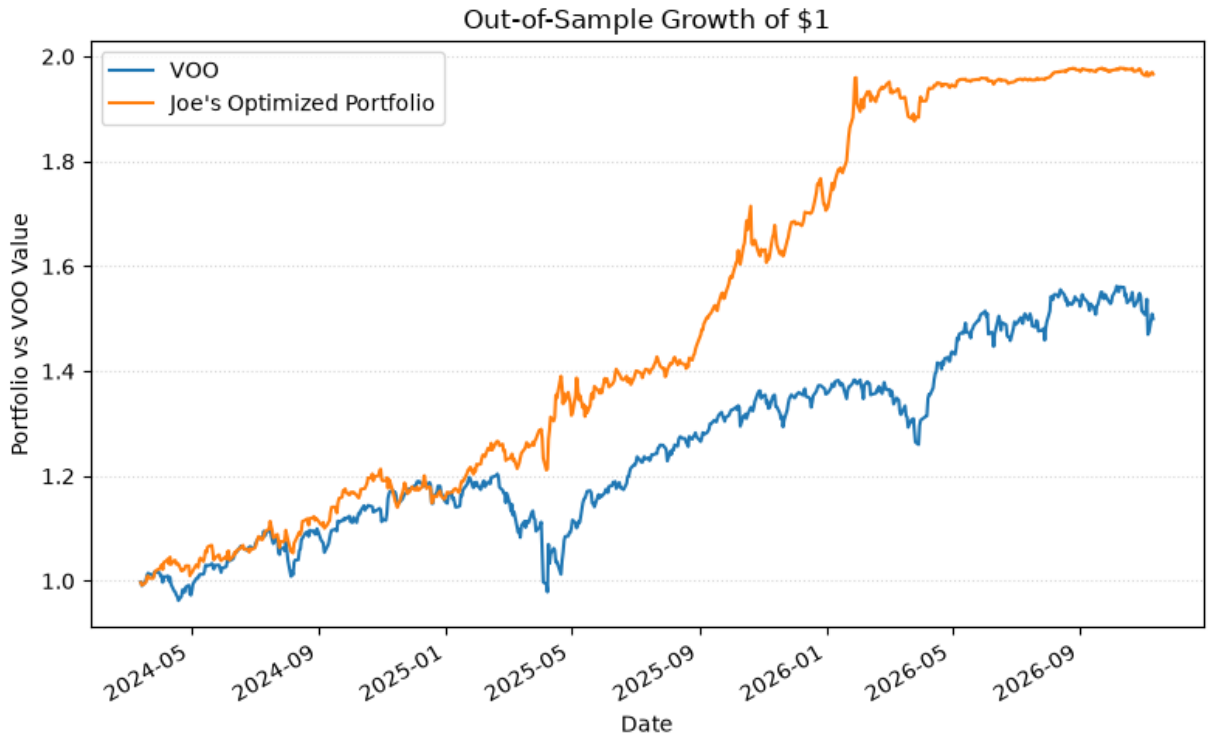


Figure 1. Out-of-sample growth of $1: long-only rolling optimized portfolio versus VOO.

The optimized portfolio and VOO began with a normalized value of $1.

Main observations

- The optimized portfolio moved ahead of VOO relatively early in the test period.
- The performance gap widened substantially during late 2025 and early 2026.
- The optimized portfolio temporarily approached approximately $1.99.
- A subsequent pullback reduced the portfolio value to approximately $1.89.
- The strategy recovered and finished at approximately $1.97.
- VOO finished at approximately $1.49 after a comparatively smoother path.
- The optimized portfolio retained a considerable cumulative-performance advantage at the end of the period.

The rise and pullback indicate that the strategy retained meaningful allocation and concentration risk even though leverage and short selling were not used.

Dynamic Allocation Analysis

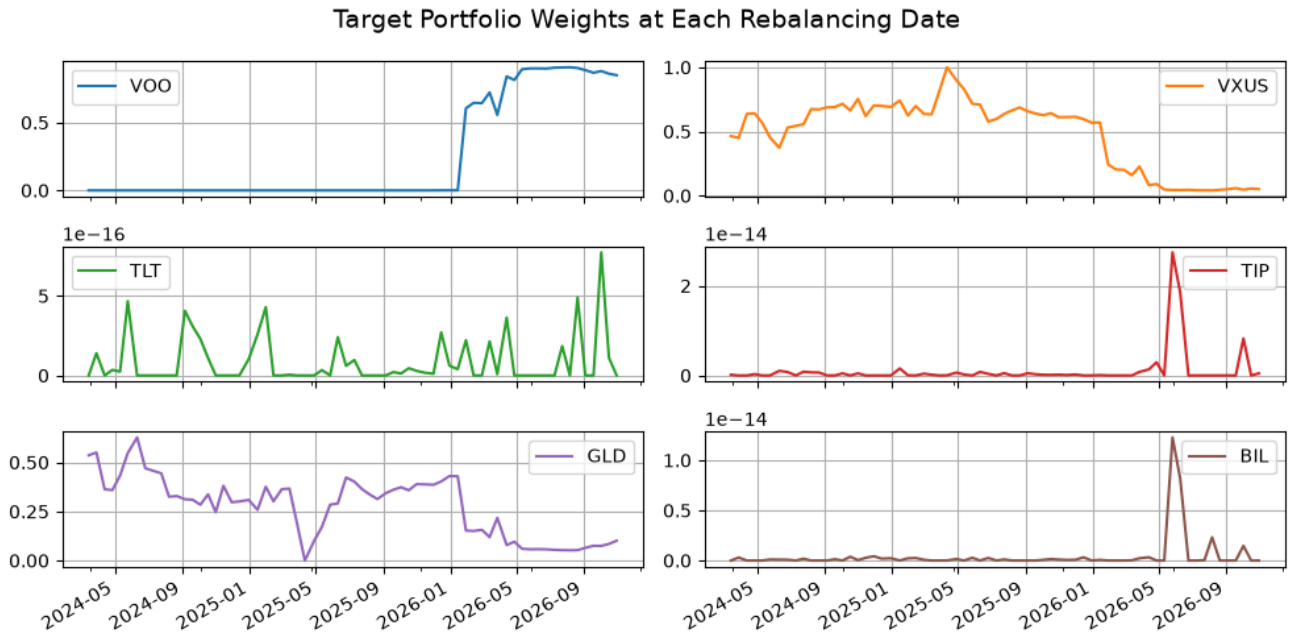


Figure 2. Long-only target asset weights at each 10-trading-day rebalancing date.

The target-weight charts show a dynamic long-only allocation rather than a stable diversified mix. The model initially favored VXUS and GLD, then shifted strongly toward VOO during 2026 while VXUS and GLD allocations declined.

VOO

VOO remained close to zero through much of the early test period. Its allocation rose sharply during 2026, reached approximately 85% to 90%, and finished near 80% to 85%.

This suggests that the optimizer initially preferred international equities and gold, then shifted toward U.S. equities when the rolling return-risk estimates became more favorable for VOO.

VXUS

VXUS received a substantial positive allocation during the first part of the test period, generally ranging from approximately 35% to 75% and briefly reaching 100%.

The VXUS weight then fell sharply during 2026 and subsequently moved close to zero. This represents a major regime change in the model's assessment of international equities.

TLT

TLT remained effectively at zero throughout the test period. The tiny values shown on the chart, around 10^-16, are numerical optimization noise rather than economically meaningful positions.

The optimizer therefore assigned no meaningful strategic role to long-term U.S. Treasuries during this test period.

TIP

TIP also remained effectively at zero. Values around 10^-14 are numerical tolerance effects and should be interpreted as a 0% allocation.

The rolling estimates did not justify a meaningful allocation to inflation-protected bonds during the test period.

GLD

GLD was one of the strategy's largest early positions. Its allocation generally ranged from approximately 25% to 60%, briefly fell near zero, and later recovered to around 30% to 40%.

The GLD allocation then fell sharply during 2026 and finished near 10%. The reduction occurred as the model shifted capital toward VOO.

This relationship should be tested formally through asset-level return-contribution analysis rather than inferred from weights alone.

BIL

BIL remained effectively at zero throughout the test period.

The tiny displayed BIL values, around 10^-14, are numerical optimization noise rather than meaningful economic exposure.

Interpretation of the Allocation Regimes

The weight history suggests two broad allocation regimes.

Regime 1: International-equity and gold allocation

The earlier period was characterized by:

- VOO exposure close to zero during most of the early period
- substantial VXUS exposure, generally around 35% to 75%
- a material GLD allocation, generally around 25% to 60%
- negligible allocations to TLT, TIP and BIL

Regime 2: Rotation toward U.S. equities

During 2026, the model shifted toward:

- a long VOO position near 90%
- VXUS exposure close to zero
- substantially lower GLD exposure
- little exposure to TIP and BIL
- negligible bond and Treasury-bill exposure

The second regime concentrated heavily in VOO, with smaller residual positions in VXUS and GLD. Because all weights were non-negative and summed to 100%, the strategy remained unleveraged.

The portfolio achieved a higher realized Sharpe ratio than VOO during the selected test period. This is encouraging evidence, but it is not sufficient to establish persistent or implementable outperformance.

Key Limitations

1. Trading costs are excluded

The backtest does not deduct:

- ETF bid-ask spreads
- brokerage costs
- market impact
- taxes or other implementation expenses

Reoptimizing every 10 trading days may generate meaningful turnover, particularly when target weights change rapidly. Net performance should therefore be tested after deducting transaction costs based on the absolute changes in portfolio weights at each rebalancing date.

2. Concentration risk remains

Although the portfolio is long-only and unleveraged, the optimizer can allocate most of the portfolio to one or two assets. During 2026, VOO received approximately 85% to 90% of the portfolio, while TLT, TIP and BIL remained effectively at zero. These boundary solutions indicate substantial concentration and sensitivity to historical expected-return estimates.

3. Limited benchmark comparison

VOO is a useful equity benchmark, but the optimized strategy can allocate across international equities, bonds, gold and Treasury bills. A balanced multi-asset benchmark would provide an additional comparison.

Additional comparisons could include:

- a 60/40 stock-bond portfolio
- an equal-weight portfolio
- a long-only maximum-Sharpe portfolio
- a minimum-variance portfolio
- a risk-parity allocation
- a volatility-matched VOO benchmark

Conclusion

The long-only rolling maximum-Sharpe strategy finished at approximately $1.97, compared with approximately $1.49 for VOO.

The results demonstrate that dynamic long-only multi-asset allocation produced materially higher cumulative wealth than VOO during the selected out-of-sample period.